

Meaningful agreement features: evidence from indexical binding*

Isabelle Charnavel, Tom Meadows & Dominique Sportiche

University of Geneva & University of California Los Angeles

1. Introduction

The analysis of *verbal agreement* in contemporary generative grammar is generally based on two assumptions. The first is that agreement represents the syntactic arguments of the inflected verb, most often the subject, effectively supplying inflectional information to the verb form. This is the logic underlying the widespread Probe-Goal architecture (Chomsky (2001)): certain functional heads bear ϕ -features whose values must be supplied or checked, and Agree supplies or checks these values from a local nominal (1).

- (1) a. $\dots [T [\phi:]] \dots [DP [\phi:1sg]] \dots$ ① The T probe searches for a Goal DP
- b. $\dots [T [\phi:1sg]] \dots [DP [\phi:1sg]] \dots$ ② ϕ -values are copied to the Probe

The second assumption is that these features, realised as agreement morphology, are not interpreted at LF, the interface to the semantic component of grammar. From the view of languages like English, in which agreement is obligatory and arguments are normally not null, this conclusion seems intuitive. Chomsky (2001) explicitly notes agreement as an exception to the Principle of Full Interpretation (Chomsky 1986), that every symbol in a syntactic representation should be interpreted at PF *and* LF. He speculates that Agree is what licenses these exceptions.

In this paper we detail a complex puzzle challenging both assumptions about agreement (see also Charnavel and Sportiche (2025) on semantic agreement). This concerns the behaviour of bound indexicals in so-called ‘Partee sentences’ (Bassi 2021), like (2)

*Thanks to Anouk Dieuleveut, David Müller, Lena Higginson, Julianne Magalhães Costa and audiences at NELS55, GLOW47, SuB28, Leibniz ZAS Berlin, Universität Vienna and Paris Lodron Universität Salzburg. The order of authors is alphabetical. This research is funded by the SNSF (grant #10001F_212936, *Indexicals, Binding and Presupposition: Towards a Typology and Theory of Bound Indexicals*, PI: Prof. Isabelle Charnavel), and in part by a UCLA Academic Senate Grant for Dominique Sportiche.

which involve light-headed relative clauses within copular sentences. We newly observe that bound indexicals can only appear in Partee sentences with *subject-relatives* (3a). Second, in a population of French speakers the agreement inside the RC needs to match the ϕ -features of the indexical to be bound (3b), as reported in (Charnavel and Sportiche 2024).

- (2) Je suis le seul [qui_i { suis / est } t_i fier de mes étudiants.]
 I be.1sg Det only.one who be.1sg be.3sg proud of my student.Pl
 ‘I am the only one who is proud of my students.’
 Free reading: Nobody else is proud of my students. *suis* ✓ / *est* ✓
 Bound reading: Nobody else is proud of their students. *suis* ✓ / *est* ✗

- (3) a. **The Subject Restriction**
 The relativised element must be a syntactic subject for indexicals to be bound.
 b. **Agreement Restriction**
 Agreement inside the RC must match the value of the bound pronouns.

We offer a unified analysis of (3) using two fundamental ingredients, *underspecification* and *licensing* (4). Relative pronouns like *qui* are binders of indexicals in French Partee sentences. But *qui* is syntactically underspecified for person and number: unless the relative pronoun *qui* is in a particular syntactic configuration it lacks the presuppositions needed to match with (and bind) the indexical inside the relative. The configuration in question is being local to ϕ -features on T, the syntactic locus subject agreement. These ϕ -features on T behave exactly as they do within nominals, as presuppositional restrictors to the domain of individuals. At LF this presupposition spreads to the underspecified subject, providing it with person/number information at this level of representation alone. Varying the featural content of T varies what information is supplied to the relative pronoun.

- (4) a. **Underspecification**
 Certain nominals are syntactically underspecified for Person.
 b. **Licensing**
 They can be licensed with a Person interpretation at LF, by being local to T.

If functional heads like T do not always acquire ϕ -features, or their values, from anywhere, the phenomenon of agreement should be rethought of as a kind of *matching requirement*, rather than information sharing. This is motivated here by Partee sentences, where the apparent controller of agreement is less specified than the agreement it controls, but is consistent with discussions in Ackema and Neeleman (2018) towards a similar conclusion.

Having reviewed the main line of argument taken in the paper, the rest of it is structured as follows. In Section 2 we review the core data of Partee sentences concerning agreement alternations and restrictions on indexical binding. In Section 3 we outline in more detail the Focus cum Binding analysis and how it interacts with a revised analysis of Agree. Outstanding issues remain which space limitations prevents us from discussing here except for a brief discussion of one of them in Section 4.

2. The Central Puzzle

Agreement in Partee Sentences: From the point of view of English and German, French appears to be much more liberal about the contexts in which local (i.e. 1st/2nd) person agreement is permitted. The environment that concerns us in this paper is a subset of relative clauses, those involved in Partee sentences. The data in (5) show that first person agreement is optional.¹ The same facts hold for second person. In this subsection we outline what does and doesn't set these constructions apart.

- (5) a. Je suis le seul qui suis/est content
I be.1sg the only who be.1sg/3sg happy
'I am the only one who is/*am happy.' **Partee Sentences**
- b. Nous sommes les seuls qui sommes/sont/*est content(s)
We be.1sg the only who be.1pl/3pl/3sg happy.Pl
'We are the only ones who are happy.'

Partee sentences are not alone in allowing local person agreement. In non-restrictive RCs modifying pronouns (6a) and cleft constructions in which pronouns are pivots (6b) local person agreement is possible RC-internally.

- (6) a. Moi, qui suis/?est content, je t' aiderai.
Me, who be.1sg/3sg happy, I you help.Fut.1sg
'I, who am happy, will help you.' **Non-restrictive RCs**
- b. C'est moi qui suis/est content
Expl.be.3sg me who be.1sg/3sg happy
'It's me who is/*am happy.' **RC in Pronominal Cleft**

To license person agreement, the syntactic environment around the relative clauses matters. If the head is not a suitable personal pronoun (as e.g. in *I, who am...*), the relative must be embedded in a particular kind of copular construction (which eventually needs to be explained). This is why local person agreement in (7a) is deviant. That the type of copular construction matters is evident from (7b), where local person agreement is out.²

- (7) a. J' ai récompensé le seul qui est/*suis travailleur
I Aux.1sg reward.Ptcp the only who be.3sg/1sg hardworking
'I rewarded the only one who is hardworking.' **No copular construction**

¹Local person in such cases is for some speakers restricted to a more formal register. Note that alternative Partee sentences can be formed using subjunctive or infinitival relatives, which some speakers judge to be more natural than those with the indicatives with local person.

²Charnavel et al. (2025), to which we refer for more detail about this point, explores the possibility that the distinction between predicational vs. specificational copular constructions (Higgins 1973, for a review see Den Dikken 2017, Heycock 2020) may matter: local person agreement is degraded where focus on the postcopular element, induced by a particular question like *What is your specificity?*, enforces a predicational construal of the copular sentence.

- b. Le seul qui est/*suis content, c'est moi
 The onlys who be.3sg/1sg happy, Expl.be.3sg me
 'The only one who is happy is me.' **Wrong kind of copular construction**

A further hurdle to local person agreement remains. Local person is only possible if the head of the relative is light (or semi-lexical) as in English, where *one* is used. Prototypical French cases involve *le seul* 'the only (one)', or superlative adjectives. If a regular lexical NP heads the relative, as in (8), local person is out.

- (8) Nous sommes les (seuls) étudiants qui *sommes/sont/*est d'accord avec
 We be.1sg Det.pl (only) student.Pl who be.1pl/3pl/3sg of agreement with
 la décision.
 Det.f decision
 'We are the (only) students who agree with the decision.'

To summarise, French permits local person agreement in relatives under specific conditions external to the relative clause itself. In the absence of personal pronominal heads, the relative clauses must modify semi-lexical/light material embedded within a copular construction containing a c-commanding element of matching person/number.

Indexical Binding in Partee Sentences: In the introduction we identified restrictions on indexical binding which we now return to. In (9), the subject RC meets all the requirements for local person agreement. The embedded possessive pronoun can have a bound reading, leading to ambiguity highlighted below the translation. At a descriptive level *mes* 'my' is behaving like a bound variable.

- (9) Je suis le seul [qui_i suis t_i fier de mes étudiants.]
 1sg be.1sg Det only who be.1sg proud of my student.Pl
 'I am the only one who is proud of my students.'
 Free reading: Nobody else is proud of my students.
Bound reading: Nobody else is proud of their students.

Once we try and widen the range of relatives in Partee sentences with the bound reading, the restriction in (10) becomes apparent. Subject relatives are not just archetypal of Partee sentences with bound indexicals, they are obligatory. This holds of speakers of English, French, and German that we have informally consulted.

- (10) **The Subject Restriction**
 An indexical bound by relativized DP within an RC can only be bound if this DP is a syntactic subject.

We can illustrate the Subject Restriction using the data in (11). These both involve non-subject relatives, in which the pronoun we are trying to bind is contained in the c-command

domain of the relative trace. In both (11a) and (11b) a bound reading is not available for the pronoun within the embedded subject.

- (11) a. Je suis le seul [que David a présenté **t** à mes étudiants
1sg be.1sg Det only that David Aux introduced to my student.Pl]
'I am the only one who David introduced to my students.'
Free reading: David didn't introduce anyone else to my students.
#Bound reading: David didn't introduce anyone else to their students.
- b. Je suis le seul [que David a informé **t** [que mes étudiants
1sg be.1sg Det only that David Aux announced that my student.Pl
étaient contents]]
were happy.Pl
'I am the only one who David informed that my students are happy.'
Free reading: David didn't inform anyone else that my students were happy.
#Bound reading: David didn't inform anyone else that their students were happy.

By syntactic subject in (10) we are referring to arguments to move to [Spec, TP]. In principle one might imagine as relevant a wider class of elements, external arguments or simply (semantic) subjects of predication. To test this we constructed examples like (12) and (13) which involve relativising the 'subject' of different kinds of small clauses. On any analysis of small clause complements, these nominals do not occupy [Spec, TP].³ In both cases we form grammatical RCs, but no bound reading is possible for pronouns in the scope of the relative trace.

- (12) Je suis le seul que David considère **t** comme un bon mentor pour mes
I be.1sg Det only who David consider.3sg as a good mentor for my
étudiants.
student.Pl
'I am the only one who David considers a good mentor for my students.'
Free reading: David considers nobody else a good mentor for my students.
Bound reading: David considers nobody else a good mentor for their students.
- (13) #Je suis le seul que David a vu **t** faire de mon mieux.
I be.1sg Det only who David Aux.3sg seen do.Inf of my best
'I am the only one who David has seen do my best.'
Bound reading: David has seen nobody else do their best.

For some French speakers, even forming Partee sentences according to Subject Restriction is not enough to guarantee the possibility of a bound reading. This is where we get to the relevance of the local person agreement. There is a population of speakers of French,

³Such relatives resemble object relatives in using the relative pronoun *que*, rather than (nominative) *qui* (Sportiche 2011).

which we will term FrenchA, for whom bound readings require agreement to match the person-number values of the pronoun to be bound.⁴ This was initially reported by Charnavel and Sportiche (2024), building on earlier observations by Ivan and Mirrazi (2019), Bassi (2021). This is the Agreement Restriction in (16).

(14) **The Agreement Restriction**

Where indexical binding obeys the Subject Restriction, the person feature value on T in the minimal clause containing the indexical ought to match that of the bound indexicals.

We can illustrate the above restriction using the data in (17) and (18). In (17) we have a subject relative with a 1SG pronoun to be bound; the bound reading is distinctly more accessible when RC-internal agreement is 1SG (17b), rather than 3SG (17a). This ‘matching effect’ also holds of bound readings for third person pronouns, as illustrated in (18). Here the bound reading on *ses* is very hard to get when agreement is 1SG (18b), compared to when agreement is 1SG.⁵

- | | | | |
|------|----|---|-----------------|
| (15) | a. | Je suis le seul [RC qui est fier de mes étudiants]
I am the only who is proud of my students
Free reading: Nobody else is proud of my students.
<u>#Bound reading:</u> Nobody else is proud of their students. | Mismatch |
| | b. | Je suis le seul [RC qui suis fier de mes étudiants]
I am the only who is proud of my students
Free reading: Nobody else is proud of my students.
<u>Bound reading:</u> Nobody else is proud of their students. | Match |
| (16) | a. | Je suis le seul [RC qui est fier de ses étudiants]
I am the only who is proud of 3sg.poss students
Free reading: Nobody else is proud of my students.
<u>Bound reading:</u> Nobody else is proud of their students. | Match |
| | b. | Je suis le seul [RC qui suis fier de ses étudiants]
I am the only who is proud of 3sg.poss students
Free reading: Nobody else is proud of my students.
<u>#Bound reading:</u> Nobody else is proud of their students. | Mismatch |

To summarise this section, indexical binding is sensitive to the morphosyntax of relative clauses in Partee sentences. English and French show a clear asymmetry between subject and non-subject relatives. More than simply requiring subject relatives, some grammars of

⁴The two francophone authors of this paper, IC and DS, are members of the French A variety.

⁵Comparing (17a) and (18b), the latter is more degraded on the bound reading than the former in FrenchA, but it should be noted that some speakers - FrenchC - allow a bound reading in (18b). They seem to be treating the property *fier de ses enfants* / proud of one's students as a ‘generic’ property applicable to any subject regardless of person marking (but a match in number is required). We do not discuss FrenchC here.

French impose a tighter restriction that needs pronouns and subject agreement to match. In the next section we provide an account of how indexical binding, subjecthood and agreement interact which requires a novel analysis of verbal agreement.

3. The Analysis

A Focus cum Binding Analysis of Bound Indexicals: To derive the availability of bound indexicals in general (i.e., also outside Partee sentences as in 17), we adopt Char-navel and Sportiche (2024)’s approach, which relies on both binding and focus, unlike previous approaches exploiting only one of these mechanisms (see Kratzer (2009) based on binding or Sauerland (2013) based on focus, i.a.).

- (17) Only I did my homework. (Heim 1991b)
Bound reading: Nobody else did their homework.

As reviewed in Char-navel and Sportiche (2024), two main types of analyses account for the bound reading in (17). Morphosyntactic approaches like Kratzer (2009) or Wurmbrand (2017a) treat *my* in (17) as a fake indexical (a pronoun without any interpreted person feature), which acquires person features at PF due to feature transmission through Agree with a functional head or a DP antecedent. Semantic approaches like Sauerland (2013) or Ja-cobson (2012) treat *my* as a real indexical, whose features need not be interpreted in focus alternatives. Both sorts of approach face empirical problems. Morphosyntactic approaches ignore the role of focus, and thus cannot explain some constraints on the specific construc-tions in which bound indexicals occur. Semantic approaches ignore the role of binding, and cannot therefore account for the role of agreement in bound indexicals in Partee sentences.

Instead, Char-navel and Sportiche (2024) follow semantic approaches in treating bound indexicals as real indexicals, i.e., as variables with presuppositional person features as in (18) (as in Heim (2008), i.a.) and in attributing a crucial role to focus. They do not, however postulate (as in Jacobson (2012)) that person features can generally be ignored in focus alternatives, nor do they derive it from pure presuppositionality (as in Sauerland (2013)), which both over- and undergenerates (as shown by McKillen (2016) i.a.). Char-navel & Sportiche instead exploit the uncontroversial (albeit implicit) assumption that the person features (and other presuppositions) of focused (F-marked) DPs are not interpreted in focus alternatives as illustrated in (19).

- (18) $\llbracket I_7 \rrbracket^{g,w,t,c} \neq \#$ iff $g(7)$ is the speaker s of c ; if $\neq \#$, $\llbracket I_7 \rrbracket^{g,w,t,c} = s_c$

- (19) a. Only $[\alpha \text{ I}_F \text{ left}]$ [i.e., no other contextually relevant individual left.]
b. $\llbracket I_7 \rrbracket^f = \{x | x \in D_e\}$ focal alternatives to I ignore the 1st person property

Here, F-marking on I implies that the alternatives can be any other individual that is contextually relevant; under Rooth (1992)’s theory, this is because the focus meaning of a

constituent (noted $\llbracket \alpha \rrbracket^f$) is a set of constituents of the same semantic type (which can be contextually restricted).

Assuming that the presuppositions of bindees and binders unify,⁶ this hypothesis directly derives the binding of indexicals. Specifically, Charnavel and Sportiche (2024) hypothesizes that identical presuppositions individually encoded on a binder and a bindee do not individually hold independently of each occurrence of the variables entering into the binding relation, but unify: this presupposition is encoded only once at LF, simultaneously restricting the range of all occurrences of this variable; derivationally, this can be implemented by the presupposition of the bindee projecting to the binder (cf. Heim and Kratzer (1998:126)’s rule of semantic agreement under lambda abstraction, to be adapted to the rule of functional application). This implies that when the binder is focused, the presupposition of all occurrences of the variable is ignored in focus alternatives as shown in (22).

- (20) Only [α I₇ saw my₇ friend].
 $\llbracket \text{my}_7 \text{ friend} \rrbracket^{g,w,t,c} = \text{the friend of } g(7) (\neq \# \text{ iff } g(7) = s_c)$
 $\llbracket 7 \text{ saw my}_7 \text{ friend} \rrbracket^{g,w,t,c} = \lambda x_e. x \text{ saw } x\text{'s friend} (\neq \# \text{ iff } x = s_c)$
 $\llbracket \text{I } 7 \text{ saw my}_7 \text{ friend} \rrbracket^{g,w,t,c} = [\lambda x_e. x \text{ saw } x\text{'s friend}](g(7)) (\neq \# \text{ iff } g(7) = s_c)$
 $\llbracket \alpha \rrbracket^o = 1 \text{ iff } s_c \text{ saw } s_c\text{'s friend}$
 $\llbracket \alpha \rrbracket^f = \{x \text{ saw } x\text{'s friend} \mid x \in D_e\}$

Returning to the case of Partee sentences, this set of hypotheses implies that the indexical in the relative clause is bound by a focused pronoun. Which pronoun is the relevant binder? One conceivable option could be that the focused binder of *mes* in (11) is the matrix subject *je* (cf. Wurmbbrand 2017); and *je* can be focused (and must be to remove the person presupposition as above). But as argued in Charnavel and Sportiche (2024), this would violate Rule H (Fox 2000) (see below examples (28)) enforces binding by the closest binder as long as the same interpretation obtains. We thus hypothesize that the relevant binder of *mes* in (11) is the relative pronoun (or its trace), which raises two questions. First, given that binding seems to require a match in featural presuppositions, how can the relative pronoun serve as an indexical binder? Second, how is focus on the relative pronoun derived? The rest of the paper answers the first question. The second question cannot be answered in detail here; we assume that focus on the relative pronoun derives from association with the head of the RC modified by adjectival *only* (cf. (Bassi 2021)).

Underspecification and agreement: In this section we detail how, at a syntactic level, local person agreement is even possible in subject relatives. This involves rethinking the analysis of agreement phenomena, which is needed on independent grounds (Charnavel and Sportiche 2025). Part of this involves an interpretative component which we defer until next subsection, but which will prove crucial to deriving the interactions with indexical binding.

⁶The present account resembles that of Stechow (2003) which assumes feature deletion at LF under semantic binding (which requires feature match). We can see the present account as providing a justification for von Stechow’s mechanism.

A probe goal model of Agreement (Chomsky 2000) assumes that agreement morphology on verbs reflects information that is (locally) supplied by a nominal. Clausal functional heads may bear features which need licensing (e.g. they lack values), and there exists an operation Agree that essentially tries to license these features from a local source: the probe searches for a Goal, where the Search operation is defined to encode a locality restriction.⁷ Applying that model to agreement optionalities like (21) will require one to posit a structural ambiguity. In particular, when the verbal inflection is 1st person, the relativized subject, which is local to the T probe, needs to bear a 1st person property.

- (21) Je suis le seul qui suis/est content
I be.1sg the only.one who be.1sg/3sg happy
'I am the only one who is/*am happy.'

Setting aside the optionality itself, it appears implausible for the relativized element to be a covert local person goal.⁸ Given that the Goal subject is relativized, we would expect to witness person properties on the relative pronoun. If such elements can bear local person properties, why is this never (to our knowledge) morphologically expressed? Languages with relative pronouns or relative markers often have them inflect for number, gender and case. A search of linguistic databases reports no clear cases of person-inflection on relative pronouns (or interrogative pronouns for that matter). In this sense, relative pronouns behave much like other nominal modifiers (e.g. demonstratives, numerals and adjectives) in showing minimal to no interaction with person (Baker 2008, den Dikken 2019).

In the cases we discuss, what is relativised is a *wh*-marked DP, realised as a relative pronoun. We thus take the lack of person marking on relative pronouns at face value: they are *wh*-DPs not syntactically specified for person (be it 1st, 2nd or 3rd). They only code number (and gender and case, in some languages). They are thus distinct from personal pronouns in not encoding person information (see fn 10 regarding third person pronouns).

- (22) [DP ... Num [NP]] **Relative pronouns lack person**

If the relativised element is not a covert person-marked goal, then how agreement works needs to be rethought. The subject in e.g. French (23) *does not supply person properties to a probe on T*. We therefore assume that T, the locus of agreement morphology in French, enters the syntactic derivation with ϕ -features already specified with values. As a result agreement is best understood in full generality as a correspondence, or compatibility, in properties between clausal functional heads and some nominal(s). The syntactic operation referred to as Agree imposes constraints on *what these correspondences apply to*, a consequence of how the Search part of Agree is characterized. For French we must assume that the Search stops at the closest DP as agreement can only be with the subject (which may of course be relativized) and nothing else.

⁷How Search is defined is controversial cf. e.g. Bjorkman and Zeijlstra (2019) and references therein.

⁸Note that it is implausible for French to analyse the gap as a silent pronoun as in (partial) null subject languages along the lines of Wurmbrand (2017) as French does not display the diagnostic properties of (partial) null subject languages.

Summarising, probes are not themselves lacking values. They search for the closest DP to pair with the probe (regardless of whether values of paired elements match). If this DP fails to match the features on the Probe that needs licensing, licensing simply fails. In the next section, we turn to what happens in such circumstances. Before doing so, note that the cases we discuss here all involve Agree failures due to a lack of person feature value match between the probe and the goal. Such failures are not limited to person features, they are also found with number features alone, as well as both number and person features at the same time. This is discussed in Charnavel and Sportiche (2025).

Presuppositional agreement Chomsky (1986) formulates a principle of economy of syntactic representation, the Principle of Full Interpretation, according to which every symbol in a syntactic representation should be interpreted at PF *and* LF. Let us accept here that inflectional ϕ -features on T are (normally) not interpreted at LF. This constitutes an exception to the principle of Full Interpretation, explicitly noted in Chomsky (2001). Let us accept with Chomsky (2001) that the agreement relation is what allows these exceptions: it allows the interfaces to ignore specific parts of syntactic representations at Spell-Out. As a corollary, this predicts that when Agree fails, the ϕ -features on T must be interpreted.

The null hypothesis is to interpretatively treat ϕ -features on T as they are normally treated in the nominal domain, as presuppositional restrictions on the domain of individuals (Heim and Kratzer 1998, Heim 2008). Quite generally, presupposition triggers constrain the interpretation of one of their arguments (a subject or a sister). For example, in the case of the adverb *again*, the presupposition bears on the interpretation of its sister node (the denoted event occurred previously); the verb *sleep* presupposes that its subject is a possible sleeper, the verb *cluster* that it is a plurality. Most relevantly, for ϕ -features, a pronoun such as say, *I*, is analysed as an individual denoting expression α argument of a person head P marked first person $P_{1P}(\alpha)$. P imposes on α its speaker denoting presupposition, namely $\llbracket 1st \rrbracket^c = \lambda x_e : x \text{ includes } s_c. x$ (the singular ϕ -feature independently imposing a singularity requirement), cf, e.g. Heim and Kratzer (1998), Heim (2008).

In the case of T,⁹ T takes (up to) two arguments, a subject and a complement, say VP. Now person marking on T will, by definition, presuppose that this property holds of an individual denoting expression and thus cannot apply to T's complement. The only element of a suitable type that is local to the presupposition trigger is the DP in [Spec, TP]. Viewing T as a binary function, and restricting ourselves to the partial function T_1 applying to the subject, such a partial function marked first person, T_1^{1st} applies to the individual-denoting subject α where $\llbracket T_1^{1st}(\alpha) \rrbracket^c$ is interpreted as including the speaker in c .¹⁰

The application of ϕ -feature triggered presupposition to the subject constrains its interpretation. If T is marked, say, [1sg], for example, the ϕ -properties of the subject will have to be compatible with this restriction on D_c . To illustrate, consider the following example:

⁹We assume here that verbal inflectional ϕ -features appear on T, but nothing hinges on this assumption. It may very well be that they mark an affixal head above T to which T head moves.

¹⁰Note that if T lacks a subject at LF, the result is ill formed as T has no suitable argument. This makes correct predictions that are discussed in Charnavel and Sportiche (2025).

- (23) *Tu suis content.
You be.1sg happy
*You am happy.

Here, Agree fails since the only DP local to T is the subject, and the features of T and this DP do not match. As a result, the features on T must be interpreted, restricting the range of the subject to the speaker. This clashes with the presupposition of *tu/you* which excludes the speaker, resulting in ill-formedness.

Why are relative pronouns, which lack a person specification, treated differently from the cases in (23)? It is essentially because semantic compatibility is a weaker requirement than matching. When clausal [1sg] imposes the requirement that the subject must denote an atomic individual including the author of the context, the relative pronoun does not have a person specification incompatible with this requirement.¹¹ Semantically, the relative pronoun is restricted to range over a set meeting the person property on T it would not otherwise range over given its underspecified syntax.

What we have outlined over 3.2 and 3.3 is an approach to agreement which is able to handle the possibility of mismatches. This account involves a division of labour between the narrow syntax and LF actually predicted by standard theories of agreement when there is a feature mismatch. This mismatch between an underspecified goal (like relative pronouns) and a probe is not a problem as long as it does not involve an interpretative conflict. We connect this view of agreement to indexical binding in next subsection.

Deriving the Subject and Agreement Restrictions: Recall that binding indexicals in French Partee sentences is subject to two restrictions, that are repeated in (26) and (27). In this section we employ the ingredients introduced in 3.1, 3.2 and 3.3 to derive these restrictions.

(24) **The Subject Restriction**

An indexical bound by relativized DP within an RC can only be bound if this DP is a syntactic subject.

(25) **The Agreement Restriction**

Where indexical binding obeys the Subject Restriction, the person feature value on T in the minimal clause containing the indexical ought to match that of the bound indexicals.

What makes indexical binding in Partee sentences more intricate is the choice of focused binder. First, the indexical external to the relative clause is blocked from binding the bound indexical directly due to the locality restriction imposed by Rule H, as we illustrate with the following two cases meant to be read with the bound reading:

¹¹We must therefore commit to analyzing third person pronouns and lexical DPs as marked, e.g. as [-participant] in a [+/- author, +/- participant] system, e.g. Harbour (2016), rather than lacking person.

- (26) a. Je suis le seul [qui_i t_i suis fier de mes étudiants.]
 1sg be.1sg Det only who be.1sg proud of my student.Pl
 ‘I am the only one who is proud of my students.’
Bound reading: Nobody else is proud of their students.
- b. Je suis le seul [que David a présenté t à mes étudiants
 1sg be.1sg Det only that David Aux introduced to my student.Pl]
 ‘I am the only one who David introduced to my students.’
 #Bound reading: David didn’t introduce anyone else to their students.

In both, the indexical *mes* could be locally bound by the trace *t*, itself bound by the subject *je* (via the relative pronoun and the head of the relative); or the indexical *mes* could be bound directly by the subject *je*, and, independently, the trace could also be bound by *je* (again via the relative pronoun and the head of the relative). As these two options yield the same interpretation, the second option is ruled out by rule H (Fox 2000). As a result, the second sentence is ruled out under the bound reading as it is subject to incompatible requirements: rule H requires the trace to be the local binder of the indexical, but the subject restriction rules this option out. The first sentence however is fine under the bound reading as both Rule H and the subject restriction can be satisfied.

We now tie together the Subject Restriction and the Agreement Restriction. In (28b), the trace *t* must bind the indexical *mes* but it (syntactically and) and semantically mismatches its feature structure, hence the person presupposition of the indexical. As semantic matching is a prerequisite for a semantic binding relation, the trace cannot bind the indexical and the bound reading is excluded. In (28a) by contrast, although not syntactically marked first person, the trace in subject position is semantically interpreted as first person, as required by the interpreted 1st person feature on T. As a result it matches the interpretation of *mes* and can therefore bind it.

These considerations derive the Subject Restriction from the Agreement Restriction: there is no Subject Restriction per se, rather, requirements on binding an indexical, say 1P, impose restrictions on the content of the binder; the binder must be interpreted at LF as 1P; it is because the subject is an argument of T and because T can carry presuppositional features of the right kind that the subject can be understood as a suitable binder: this binding requirement derives the Agreement Restriction.

4. Further issues: when agreement doesn’t matter

It is fairly obvious that Agreement Restriction does not, at superficial level at least, universally hold. English does not allow agreement alternations in Partee sentences, yet indexical binding is unaffected. This resembles the behaviour of French speakers in the context of infinitival relatives like (27), which do not display any verbal agreement. Why does agreement only matter to indexical binding in some subject RCs and not others?

- (27) Je suis le seul [RC à être fier de **mes** étudiants]
 I am the only to be proud of my students

Free reading: Nobody else is proud of my students.

✓Bound reading: Nobody else is proud of their students.

A natural suggestion would be to invoke competition: since the subject of the relative must bind a first person indexical, encoding of a first person property on this subject is required when possible: this would mandate first person marking on T in (17) in case of the bound reading, and would allow indexical binding in English and in infinitives since person marking is not an option. In effect, this would capitalize on Heim's 1991a *Maximize Presupposition* Principle. When there is a single agreement possible, as in English, or infinitival relatives where no exponence of agreement is visible, binding by a subject of an indexical (yielding a bound reading) is possible.

The central difficulty with this otherwise plausible approach is the subject restriction: since there is no option for person marking on non subject relativisees, this account incorrectly predicts that there should be no subject restriction at all since the possible presupposition encoding would always be maximal.

An alternative analysis could take the following form. Even in these infinitival relatives and quite generally in English, there is a person specification on T that sets subjects apart from non subjects. Instead, what is allowed in these languages or constructions is impoverished morphophonological realisation of the person marking. It may in fact be that some such assumption is needed anyway, at least for the case of tensed relative clauses or clefts in which there can be a person mismatch between the head of the relative clause or the focus of the cleft and the wh-clause predicted of it.

References

- Ackema, Peter, and Ad Neeleman. 2018. *Features of person: From the inventory of persons to their morphological realization*, volume 78. MIT Press.
- Baker, Mark. 2008. *The Syntax of Agreement and Concord*. Cambridge: Cambridge University Press.
- Bassi, Itai. 2021. Fake features and valuation from context. Doctoral dissertation, MIT.
- Bjorkman, Bronwyn M, and Hedde Zeijlstra. 2019. Checking up on (ϕ -) agree. *Linguistic Inquiry* 50:527–569.
- Charnavel, Isabelle, Anouk Dieuleveut, David Müller, Tom Meadows, and Dominique Sportiche. 2025. Who am I to (dis)agree? Interpretation sensitive agreement: Experimental evidence from french relatives. Presented at SALT 35, Harvard University, Cambridge, Massachusetts.
- Charnavel, Isabelle, and Dominique Sportiche. 2024. Indexical binding, presuppositions and agreement. In *Proceedings of Sinn und Bedeutung 2023*. <https://ling.auf.net/lingbuzz/007872>.
- Charnavel, Isabelle, and Dominique Sportiche. 2025. On semantic agreement. Unpublished ms. Université de Genève and UCLA.
- Chomsky, Noam. 1986. *Knowledge of language*. New York, New York: Praeger Publishers.

- Chomsky, Noam. 2000. Minimalist Inquiries: The framework. In *Step by step: Essays on minimalist syntax in honor of Howard Lasnik*, ed. by Roger Martin, David Michaels, and Juan Uriagereka, 89–156. Cambridge, MA: MIT Press.
- Chomsky, Noam. 2001. Derivation by phase. In *Ken Hale: A life in language*, ed. by Michael Kenstowicz, 1–52. Cambridge, MA: MIT Press.
- den Dikken, Marcel. 2019. The Attractions of Agreement: Why Person is Different. *Frontiers in Psychology*.
- Fox, Danny. 2000. *Economy and semantic interpretation*. Cambridge, Massachusetts: MIT Press.
- Harbour, Daniel. 2016. *Impossible persons*, volume 74. MIT Press.
- Heim, Irene. 1991a. Artikel und definitheit. In *Semantik: Ein internationales Handbuch der zeitgenössischen Forschung*, 487–535. Berlin: Mouton de Gruyter.
- Heim, Irene. 1991b. *Class Notes on Control*. MIT: MIT.
- Heim, Irene. 2008. Features on bound pronouns. In *Phi theory*, ed. by Daniel Harbour, David Adger, and Susana Béjar, 35–56. Oxford: Oxford University Press.
- Heim, Irene, and Angelika Kratzer. 1998. *Semantics in generative grammar*. Malden, MA: Blackwell.
- Ivan, Rodica, and Zahra Mirrazi. 2019. Minimal pronouns & T-Agreement effects: the case of Farsi Fake Indexicals. In *Talk given at GLOW42, Oslo*.
- Jacobson, Pauline. 2012. The direct compositionality and “Uninterpretability”: The case of (sometimes) “Uninterpretable” features on pronouns. *Journal of Semantics* 1–39.
- Kratzer, Angelika. 2009. Making a pronoun: fake indexicals as windows into the properties of pronouns. *Linguistic Inquiry* 40:187–237.
- McKillen, Alanah. 2016. On the Interpretation of Reflexive Pronouns. Doctoral dissertation, McGill.
- Rooth, Mats. 1992. A theory of focus interpretation. *Journal of Semantics* 1:75–116.
- Sauerland, Uli. 2013. Presuppositions and the alternative tier. In *Proceedings of SALT23*, ed. by Todd Snider, 156–173.
- Stechow, Arnim von. 2003. Feature deletion under semantic binding. In *Proceedings of the North East Linguistics Society*, ed. by Makoto Kadowaki and Shigeto Kawahara, volume 33, 377–403. Graduate Linguistic Student Association, Amherst, Massachusetts: University of Massachusetts at Amherst.
- Wurmbrand, Susi. 2017. Icelandic as a partial null subject language: Evidence from fake indexicals. *Order and structure in syntax II* 339.
- Wurmbrand, Susi. 2017a. Feature sharing or how I value my son. In *The Pesky Set: Papers for David Pesetsky*, ed. by Claire Halpert, Hadas Kotek, and Coppe van Urk, 173–182.

Isabelle Charnavel, Tom Meadows & Dominique Sportiche

isabelle.charnavel@unige.ch, tom.meadows@unige.ch, dominique.sportiche@ucla.edu